

Qualification Pack



Automated Metal Cutting Plasma Engineer

QP Code: CSC/Q0216

Version: 1.0

NSQF Level: 6

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CSC/Q0216: Automated Metal Cutting Plasma Engineer

Brief Job Description

An Automated Metal Cutting Plasma Engineer is a professional who specializes in using advanced machines that cut metal using high-temperature plasma.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CSC/N0221: Oversee the Setup of Automated Metal Cutting Plasma.](#)
2. [CSC/N0222: Select and implement automated CNC plasma cutting techniques for metal fabrication.](#)
3. [CSC/N0223: Post-Cutting Quality Check and Finishing](#)
4. [CSC/N1339: Collaboratively coordinate with the team](#)
5. [CSC/N0505: Follow health, safety and environment guidelines at workplace](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Machining, Welding, Manufacturing
Occupation	Welding and Cutting
Country	India
NSQF Level	6
Credits	22

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7412.0101
Minimum Educational Qualification & Experience	Completed 4 year UG program (Mechanical/Automobile/Electrical/Electronics) OR Completed 2nd year diploma after 12th with 3 Years of experience relevant OR Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience relevant OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience relevant
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-06-IT-04200-2025-V1-CGSC
NQR Version	1.0

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CSC/N0221: Oversee the Setup of Automated Metal Cutting Plasma.

Description

This NOS unit is about to Oversee the Setup of Automated Metal Cutting Plasma

Scope

The scope covers the following :

- Perform automated metal cutting plasma by inspecting equipment, preparing materials, and configuring cutting programs for efficient operation.
- Set up Gas supply to the plasma cutting machine & do the alignment.
- Prepare and set up automated metal cutting plasma systems with a focus on safety measures, test cuts, and quality control for optimal operation.

Elements and Performance Criteria

Perform automated metal cutting plasma by inspecting equipment, preparing materials, and configuring cutting programs for efficient operation.

To be competent, the user/individual on the job must be able to:

- PC1.** Inspect the plasma cutting machine, ensuring all components are in good condition and properly functioning. This includes the cutting torch, power supply, CNC controller, gas supply, and any safety features.
- PC2.** Prepare the metal sheets or work pieces to be cut. This involves ensuring they are clean, free from any contaminants, and properly positioned on the cutting table or fixture.
- PC3.** Load the cutting program into the CNC controller or computer interface. Verify that the program includes the correct cutting parameters such as cutting speed, amperage, gas flow rates, and pierce delay.

Set up Gas supply to the plasma cutting machine & do the alignment

To be competent, the user/individual on the job must be able to:

- PC4.** Select the appropriate gas for the cutting process based on the material being cut. Common gases used include oxygen, nitrogen, and argon, each offering different advantages depending on the application
- PC5.** Set the gas pressure and flow rates according to the manufacturer's recommendations and the requirements of the cutting program. Proper gas control is crucial for achieving clean and accurate cuts.
- PC6.** Ensure the cutting torch is properly positioned and aligned with the starting point of the cut on the work piece. Proper alignment helps to prevent errors in the cutting path and ensures consistent quality.

Prepare and set up automated metal cutting plasma systems with a focus on safety measures, test cuts, and quality control for optimal operation

To be competent, the user/individual on the job must be able to:

- PC7.** Implement safety measures to protect personnel and equipment during operation. This includes wearing appropriate personal protective equipment (PPE), installing safety guards, and ensuring emergency stop buttons are easily accessible.

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- PC8.** Conduct test cuts on scrap material to verify the cutting parameters and make any necessary adjustments before proceeding with the actual cutting job.
- PC9.** • Monitor the cutting process closely to ensure the desired cut quality is achieved. This involves checking for smooth edges, minimal dross (residue), and accurate dimensions according to the specification.
-

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Equipment Familiarity: Understanding the components of a plasma cutting system is crucial. This includes the cutting torch, power supply, CNC controller, gas supply system, and any additional accessories like height control mechanisms.
- KU2.** Material Characteristics: Knowledge of different metals and their properties is necessary to select appropriate cutting parameters. Factors such as thickness, conductivity, and composition influence the cutting process.
- KU3.** Software Proficiency: Familiarity with CNC programming and operation software is essential for loading cutting programs, adjusting parameters, and troubleshooting any software-related issues.
- KU4.** Gas Dynamics: Understanding the role of gases in the plasma cutting process is vital. Different gases (e.g., oxygen, nitrogen) are used for different materials and cutting requirements. Knowledge of gas pressure, flow rates, and gas interactions with the plasma arc is essential.
- KU5.** Safety Protocols: Knowledge of safety procedures and precautions is critical to prevent accidents and injuries. This includes understanding potential hazards associated with plasma cutting, wearing appropriate personal protective equipment (PPE), and implementing safety features on the equipment.
- KU6.** Troubleshooting Skills: Being able to identify and address common issues that may arise during setup and operation, such as arc instability, consumable wear, or material warping, is essential for maintaining productivity and quality.
- KU7.** Quality Control Techniques: Understanding how to evaluate cut quality through visual inspection, dimensional measurements, and other quality control methods ensures that finished parts meet specifications.
- KU8.** Maintenance Practices: Knowledge of routine maintenance tasks, such as cleaning, lubrication, and replacement of consumable parts, helps ensure the longevity and reliability of the plasma cutting system.
- KU9.** Environmental Considerations: Awareness of environmental factors such as ventilation requirements, noise levels, and proper waste disposal methods is necessary to create a safe and compliant workspace.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them

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- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform automated metal cutting plasma by inspecting equipment, preparing materials, and configuring cutting programs for efficient operation.</i>	10	10	-	6
PC1. Inspect the plasma cutting machine, ensuring all components are in good condition and properly functioning. This includes the cutting torch, power supply, CNC controller, gas supply, and any safety features.	3	2	-	2
PC2. Prepare the metal sheets or work pieces to be cut. This involves ensuring they are clean, free from any contaminants, and properly positioned on the cutting table or fixture.	3	4	-	2
PC3. Load the cutting program into the CNC controller or computer interface. Verify that the program includes the correct cutting parameters such as cutting speed, amperage, gas flow rates, and pierce delay.	4	4	-	2
<i>Set up Gas supply to the plasma cutting machine & do the alignment</i>	15	15	-	10
PC4. Select the appropriate gas for the cutting process based on the material being cut. Common gases used include oxygen, nitrogen, and argon, each offering different advantages depending on the application	5	5	-	3
PC5. Set the gas pressure and flow rates according to the manufacturer's recommendations and the requirements of the cutting program. Proper gas control is crucial for achieving clean and accurate cuts.	5	5	-	3
PC6. Ensure the cutting torch is properly positioned and aligned with the starting point of the cut on the work piece. Proper alignment helps to prevent errors in the cutting path and ensures consistent quality.	5	5	-	4
<i>Prepare and set up automated metal cutting plasma systems with a focus on safety measures, test cuts, and quality control for optimal operation</i>	15	14	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC7. Implement safety measures to protect personnel and equipment during operation. This includes wearing appropriate personal protective equipment (PPE), installing safety guards, and ensuring emergency stop buttons are easily accessible.	5	4	-	2
PC8. Conduct test cuts on scrap material to verify the cutting parameters and make any necessary adjustments before proceeding with the actual cutting job.	6	6	-	1
PC9. Monitor the cutting process closely to ensure the desired cut quality is achieved. This involves checking for smooth edges, minimal dross (residue), and accurate dimensions according to the specification.	4	4	-	2
NOS Total	40	39	-	21

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0221
NOS Name	Oversee the Setup of Automated Metal Cutting Plasma.
Sector	Capital Goods
Sub-Sector	
Occupation	Welding and Cutting
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CSC/N0222: Select and implement automated CNC plasma cutting techniques for metal fabrication.

Description

This unit is about to Select and implement automated CNC plasma cutting techniques for metal fabrication.

Scope

The scope covers the following :

- Select and execute automated metal cutting plasma techniques with precision, efficiency, and quality assurance.
- Evaluate productivity and accuracy considering machine automation
- Maximize efficiency and precision in automated metal cutting plasma techniques through advanced software utilization while minimizing environmental impact

Elements and Performance Criteria

Select and execute automated metal cutting plasma techniques with precision, efficiency, and quality assurance.

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure the technique selected is appropriate for the specific material being cut.
- PC2.** Consider the thickness of the metal to be cut and choose a technique that can handle it efficiently.
- PC3.** Ensure desired quality of the cut influences the selection of the plasma cutting technique.

Evaluate productivity and accuracy considering machine automation

To be competent, the user/individual on the job must be able to:

- PC4.** Efficiency is crucial in industrial settings. Evaluate the cutting speed and productivity of the technique to ensure it meets production demands while maintaining quality standards.
- PC5.** Precision is vital, especially for intricate designs or parts requiring tight tolerances. Look for techniques that offer high precision and accuracy in cutting.
- PC6.** Consider the level of automation required based on production volume, complexity of cuts, and available budget.

Maximize efficiency and precision in automated metal cutting plasma techniques through advanced software utilization while minimizing environmental impact.

To be competent, the user/individual on the job must be able to:

- PC7.** Choose systems with user-friendly interfaces and advanced features for efficient operation.
- PC8.** Evaluate the longevity of consumables and the ease of maintenance when selecting a plasma cutting technique.
- PC9.** Choose techniques that minimize environmental footprint while maintaining performance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Principles of Plasma Cutting: They should comprehend the fundamental principles behind plasma cutting, including how plasma is generated, its interaction with metal, and the factors influencing cut quality and efficiency.
- KU2.** Equipment Knowledge: An individual should be familiar with the components and operation of plasma cutting equipment, including plasma torches, power sources, gas supply systems, and CNC or robotic controls
- KU3.** Material Characteristics: Understanding the properties of different metals, such as conductivity, thickness, and composition, is crucial for selecting appropriate cutting parameters and achieving desired results.
- KU4.** Safety Procedures: Prioritizing safety is paramount. They must be well-versed in safety protocols related to handling equipment, operating machinery, and mitigating potential hazards like electrical shock, arc flash, fumes, and fire.
- KU5.** Setup and Calibration: Proper setup and calibration of the plasma cutting system are essential for optimal performance. This involves adjusting parameters such as gas flow rates, arc voltage, cutting speed, and pierce height based on the material being cut.
- KU6.** Programming Skills: If working with CNC or robotic systems, they should have programming skills to create or modify cutting paths, import CAD files, and generate toolpaths using software interfaces.
- KU7.** Quality Control Techniques: They need to understand factors affecting cut quality, such as dross formation, kerf width, and edge smoothness, and employ quality control techniques to ensure the final product meets specifications.
- KU8.** Environmental Considerations: Being aware of environmental factors associated with plasma cutting, such as fume extraction and waste disposal, is important for minimizing the impact on health and the environment.
- KU9.** Continuous Learning and Improvement: Plasma cutting technology is continually evolving, so individuals should stay updated on advancements, new techniques, and best practices through training programs, industry publications, and networking opportunities.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Select and execute automated metal cutting plasma techniques with precision, efficiency, and quality assurance.</i>	10	10	-	6
PC1. Ensure the technique selected is appropriate for the specific material being cut.	3	2	-	2
PC2. Consider the thickness of the metal to be cut and choose a technique that can handle it efficiently.	3	4	-	2
PC3. Ensure desired quality of the cut influences the selection of the plasma cutting technique.	4	4	-	2
<i>Evaluate productivity and accuracy considering machine automation</i>	15	15	-	10
PC4. Efficiency is crucial in industrial settings. Evaluate the cutting speed and productivity of the technique to ensure it meets production demands while maintaining quality standards.	5	5	-	3
PC5. Precision is vital, especially for intricate designs or parts requiring tight tolerances. Look for techniques that offer high precision and accuracy in cutting.	5	5	-	3
PC6. Consider the level of automation required based on production volume, complexity of cuts, and available budget.	5	5	-	4
<i>Maximize efficiency and precision in automated metal cutting plasma techniques through advanced software utilization while minimizing environmental impact.</i>	15	15	-	4
PC7. Choose systems with user-friendly interfaces and advanced features for efficient operation.	6	6	-	2
PC8. Evaluate the longevity of consumables and the ease of maintenance when selecting a plasma cutting technique.	5	4	-	1
PC9. Choose techniques that minimize environmental footprint while maintaining performance	4	5	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	40	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0222
NOS Name	Select and implement automated CNC plasma cutting techniques for metal fabrication.
Sector	Capital Goods
Sub-Sector	
Occupation	Welding and Cutting
NSQF Level	6
Credits	6
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CSC/N0223: Post-Cutting Quality Check and Finishing

Description

This unit is about to Perform Post-Cutting Quality Check and Finishing.

Scope

The scope covers the following :

- Verifying dimensional accuracy, edge quality, and conformity to specifications for quality assurance.
- Ensure Finishing processes, including deburring and surface treatment, stem.
- Ensure Quality metrics and continuous improvement

Elements and Performance Criteria

Verify dimensional accuracy, edge quality, and conformity to specifications for quality assurance. networking.

To be competent, the user/individual on the job must be able to:

- PC1.** Check for defects like dross, slag, and rough edges.
- PC2.** Verify measurements align with specifications using precise tools.
- PC3.** Assess roughness and integrity of the cut surface.

Ensure Finishing processes, including deburring and surface treatment, stem.

To be competent, the user/individual on the job must be able to:

- PC4.** Ensure edges are smooth and free from burrs.
- PC5.** Confirm actual thickness matches specifications.
- PC6.** Check angles, radii, and other geometric features

Ensure Quality metrics and continuous improvement

To be competent, the user/individual on the job must be able to:

- PC7.** Apply coatings or treatments uniformly as specified
- PC8.** Ensure adequate measures are taken to prevent corrosion
- PC9.** Maintain accurate records of inspection results and processes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Various inspection techniques and tools such as visual inspection, dimensional measurement tools, and surface roughness testers
- KU2.** The importance of adhering to dimensional tolerances and specifications to meet customer requirements
- KU3.** Different surface treatment methods like painting, powder coating, plating, or anodizing, and their impact on corrosion resistance and appearance.

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- KU4.** Safety protocols to ensure the well-being of personnel during inspection and finishing processes, including proper handling of tools and chemicals.
- KU5.** The role of documentation in maintaining quality control records and traceability throughout the production process
- KU6.** Continuous improvement strategies to optimize post-cutting inspection and finishing processes, minimize defects, and enhance efficiency.
- KU7.** Collaboration with other departments such as design, production, and quality assurance to address any issues and improve overall product quality.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** apply problem-solving approaches to different situations
- GS7.** analyse the business impact and disseminate relevant information to others
- GS8.** apply balanced judgments to different situations
- GS9.** check the work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Verify dimensional accuracy, edge quality, and conformity to specifications for quality assurance. networking.</i>	10	10	-	7
PC1. Check for defects like dross, slag, and rough edges.	3	3	-	2
PC2. Verify measurements align with specifications using precise tools.	3	3	-	2
PC3. Assess roughness and integrity of the cut surface.	4	4	-	3
<i>Ensure Finishing processes, including deburring and surface treatment, stem.</i>	15	15	-	9
PC4. Ensure edges are smooth and free from burrs.	5	5	-	3
PC5. Confirm actual thickness matches specifications.	5	5	-	3
PC6. Check angles, radii, and other geometric features	5	5	-	3
<i>Ensure Quality metrics and continuous improvement</i>	15	15	-	4
PC7. Apply coatings or treatments uniformly as specified	5	5	-	2
PC8. Ensure adequate measures are taken to prevent corrosion	5	5	-	1
PC9. Maintain accurate records of inspection results and processes	5	5	-	1
NOS Total	40	40	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0223
NOS Name	Post-Cutting Quality Check and Finishing
Sector	Capital Goods
Sub-Sector	
Occupation	Welding and Cutting
NSQF Level	6
Credits	7
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CSC/N1339: Collaboratively coordinate with the team

Description

This OS unit is about building relationships and working with people and groups inside and outside the organization, using skills and habits, to achieve the team goals and objectives

Scope

The scope covers the following :

- This unit/task covers the following:
- Creating team environment
- Communicating - giving and receiving
- Working cooperatively
- Participating in team decision making
- Demonstrating Sense of Responsibility
- Showing respect for opinions, customs, and preferences

Elements and Performance Criteria

Communicate effectively at the workplace

To be competent, the user/individual on the job must be able to:

- PC1.** exchange information and instruction with colleagues, and seek clarifications and feedback
- PC2.** assist colleagues where required
- PC3.** follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person)
- PC4.** document and share all relevant information with stakeholders in agreed formats and as per agreed timelines

Work effectively

To be competent, the user/individual on the job must be able to:

- PC5.** identify and obtain clarity regarding organisational, team and own goals and targets
- PC6.** prioritise and plan work in order to achieve goals and targets
- PC7.** monitor own and team performance as per agreed plan
- PC8.** complete duties accurately, systematically and within required timeframes
- PC9.** express emotions appropriately at the workplace and manage own response to heightened emotions
- PC10.** maintain orderliness and cleanliness in the work area Maintain and enhance professional competence
- PC11.** identify own strengths and weaknesses in relation to goals and targets
- PC12.** adapt self, service, or product to meet success criteria
- PC13.** seek and select opportunities for continuous professional development
- PC14.** formulate a professional development plan to enhance capabilities

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- PC15.** build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations
- PC16.** examine developments and trends in field of work and their potential impact on work
- PC17.** take feedback from peers, supervisors and clients to improve own performance and practices
Work in a disciplined and ethical manner

To be competent, the user/individual on the job must be able to:

- PC18.** perform tasks as per workplace standards, organizational policies and legislative requirements
- PC19.** display appropriate professional appearance at the workplace and adhere to the organizational dress code
- PC20.** demonstrate responsible and disciplined behavior at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behavior at all times, adopting environment- friendly practices, etc.
- PC21.** identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution
- PC22.** protect the rights of the client and organization when delivering services
- PC23.** ensure services are delivered equally to all clients regardless of personal and cultural beliefs
- PC24.** operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities
- PC25.** follow organizational guidelines and legal requirements on disclosure and confidentiality

Uphold social diversity at the workplace

To be competent, the user/individual on the job must be able to:

- PC26.** recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes
- PC27.** identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace
- PC28.** use inclusive or neutral language and gestures in all interactions
- PC29.** respect the personal and professional space of others
- PC30.** access grievance redressal mechanisms as per legislations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organisation's policies and procedures for working with colleagues, roles and responsibilities
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** different methods of communication and the circumstances in which it is appropriate to use these
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** different types of information that colleagues might need and the importance of providing this information when it is required

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KU7. the importance of helping colleagues with problems, to meet quality and time standards as a team

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write instructions, guidelines, procedures, messages, emails, and other media in language of the workplace
- GS2.** communicate in common and technical terms in language of the workplace
- GS3.** listen effectively and orally communicate information
- GS4.** be punctual, do work scheduling and reporting
- GS5.** comply with workplace practices and ethics
- GS6.** maintain cleanliness and healthy environment
- GS7.** be customer friendly - understand real needs of the customer and suggest most appropriate solution
- GS8.** be safety conscious and avoid risk
- GS9.** be observant, vigilant, and security consciousness
- GS10.** respond, handle problem, and escalate as necessary
- GS11.** ask for clarification and advice from concerned persons
- GS12.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS13.** plan and organize work to achieve targets and deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively at the workplace</i>	7	20	-	-
PC1. exchange information and instruction with colleagues, and seek clarifications and feedback	-	-	-	-
PC2. assist colleagues where required	-	-	-	-
PC3. follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person)	-	-	-	-
PC4. document and share all relevant information with stakeholders in agreed formats and as per agreed timelines	-	-	-	-
<i>Work effectively</i>	7	20	-	-
PC5. identify and obtain clarity regarding organisational, team and own goals and targets	-	-	-	-
PC6. prioritise and plan work in order to achieve goals and targets	-	-	-	-
PC7. monitor own and team performance as per agreed plan	-	-	-	-
PC8. complete duties accurately, systematically and within required timeframes	-	-	-	-
PC9. express emotions appropriately at the workplace and manage own response to heightened emotions	-	-	-	-
PC10. maintain orderliness and cleanliness in the work area Maintain and enhance professional competence	-	-	-	-
PC11. identify own strengths and weaknesses in relation to goals and targets	-	-	-	-
PC12. adapt self, service, or product to meet success criteria	-	-	-	-
PC13. seek and select opportunities for continuous professional development	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. formulate a professional development plan to enhance capabilities	-	-	-	-
PC15. build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations	-	-	-	-
PC16. examine developments and trends in field of work and their potential impact on work	-	-	-	-
PC17. take feedback from peers, supervisors and clients to improve own performance and practices	-	-	-	-
<i>Work in a disciplined and ethical manner</i>	8	20	-	-
PC18. perform tasks as per workplace standards, organizational policies and legislative requirements	-	-	-	-
PC19. display appropriate professional appearance at the workplace and adhere to the organizational dress code	-	-	-	-
PC20. demonstrate responsible and disciplined behavior at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behavior at all times, adopting environment- friendly practices, etc.	-	-	-	-
PC21. identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution	-	-	-	-
PC22. protect the rights of the client and organization when delivering services	-	-	-	-
PC23. ensure services are delivered equally to all clients regardless of personal and cultural beliefs	-	-	-	-
PC24. operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities	-	-	-	-
PC25. follow organizational guidelines and legal requirements on disclosure and confidentiality	-	-	-	-
<i>Uphold social diversity at the workplace</i>	8	10	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes	-	-	-	-
PC27. identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace	-	-	-	-
PC28. use inclusive or neutral language and gestures in all interactions	-	-	-	-
PC29. respect the personal and professional space of others	-	-	-	-
PC30. access grievance redressal mechanisms as per legislations	-	-	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N1339
NOS Name	Collaboratively coordinate with the team
Sector	Capital Goods
Sub-Sector	Generic
Occupation	Generic
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	01/10/2025
Next Review Date	01/10/2030
NSQC Clearance Date	01/10/2025

Qualification Pack

CSC/N0505: Follow health, safety and environment guidelines at workplace

Description

This OS unit is about following adequate safety procedures to make work environment healthy and safe

Scope

The scope covers the following :

- This unit/task covers the following:
- Adhere to standard safety procedures of the company
- Follow healthy practices and posture
- Practice waste management and recycling
- Conserve material and resources

Elements and Performance Criteria

Adhere to standard safety procedures of the organisation

To be competent, the user/individual on the job must be able to:

- PC1.** comply with general safety procedures and those for handling equipment, tools, chemicals, and hazardous material, as prescribed and followed in the organisation
- PC2.** remove finger rings or any other metal objects likely to interfere with the work
- PC3.** ensure that identification badge or any other object worn around the neck or on the clothing does not get caught in any rotating machine, or otherwise interfere with the work
- PC4.** use appropriate safety devices such as goggles, gloves, ear plugs, caps, ESD pins, covers, shoes, helmets etc. recommended for the work being performed
- PC5.** inform, escalate, or raise alarm about any suspicions, unaccounted hazardous material, devices, or other objects found in the premises
- PC6.** inform, escalate, or raise alarm about any breach of safety or security procedure in the organisation
- PC7.** help achieve zero accidents goals at work
- PC8.** avoid damage to sensitive electronic components due to negligence of ESD procedures
- PC9.** participate regularly in fire drills or other safety related workshops organised by the organisation
- PC10.** follow strictly all access control and perimeter safety procedures in designated factory areas such as robotic work stations, automated production lines, automated material movement and other potentially risky operations
- PC11.** ensure that other people follow all access control and perimeter safety procedures in designated factory areas and help avoid accidents
- PC12.** use emergency switches or other mechanisms of stopping a machine immediately in case any emergency situation has developed or about to happen
- PC13.** ensure that electrical equipment are properly grounded
- PC14.** follow Cyber Security guidelines and be vigilant at workplace

Qualification Pack

PC15. proceed to designated safe assembly area immediately on hearing fire alarm

Follow healthy practices and posture

To be competent, the user/individual on the job must be able to:

PC16. wash hands and use sanitizers as recommended to prevent spread of diseases

PC17. follow common personal hygiene practices

PC18. maintain appropriate posture, especially in long hours of sitting or standing position and in handling heavy materials

PC19. participate in company organised health sessions such as exercises, games, yoga, physiotherapy, and other activities

PC20. handle heavy and hazardous materials with care, while maintaining appropriate posture, using suitable tools, and handling equipment such as trolleys, jacks, and ladders

PC21. learn and apply first aid devices available in the workplace

PC22. learn and apply safety and handling procedures for electrical shock and electrocution

PC23. learn and apply emergency medical help services

PC24. follow workplace decorum and avoid emotional outbursts or inappropriate language

PC25. prevent any harassment at workplace

Practice waste management and recycling

To be competent, the user/individual on the job must be able to:

PC26. identify recyclable, non-recyclable, and hazardous waste generated in the workplace and comply with their disposal procedures

PC27. dispose non-recyclable waste and hazardous waste following recommended processes

PC28. deposit recyclable and reusable material at identified locations

PC29. support education and compliance of waste management processes

Conserve material and resources

To be competent, the user/individual on the job must be able to:

PC30. identify ways to optimize usage of material and resources such as water, electricity, energy in various tasks, activities, and processes

PC31. check for spills and leakages of material in various tasks, activities, and processes and plug them

PC32. escalate the leakage issue to appropriate authority if needed

PC33. carry out routine cleaning of tools, machines, and equipment and maintain them in good working condition to optimize efficiency and wastage

PC34. check if the equipment is functioning normally before commencing work and rectify or report any malfunctioning to the responsible agency

PC35. check for any odour, sparks, fumes, emission, unusual vibration, noise, or any other objectionable presence in the environment and take immediate corrective action followed by report to responsible agency

PC36. ensure electrical equipment are properly connected for use and are switched off when not in use

PC37. support education and compliance of resource conservation processes

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** company policies on workplace, environment, and personnel management
- KU2.** company policy on occupational safety and health
- KU3.** professional hazards related to nature of work and how to deal with them
- KU4.** how to maintain the work area safe and secure
- KU5.** how to handle hazardous materials, tools, and equipment
- KU6.** emergency procedures for fire, electrocution, physical injury, wounds, etc.
- KU7.** need for proper body posture and use of appropriate handling equipment
- KU8.** understand electrical grounding practices
- KU9.** common sources of pollution and ways to minimize it
- KU10.** waste management - categorisation, colour coding, handling, and disposal procedure
- KU11.** organisation policies and procedures for minimizing waste
- KU12.** efficient use of electricity, material, and water in processes
- KU13.** organization policies regarding network usage and security
- KU14.** norms for professional behaviour at workplace and dealing with deviations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicating in the language of the workplace
- GS2.** reading and interpreting documents, drawings, symbols, and instructions
- GS3.** operating computer and common office equipment and diagnosing common electrical and interconnection problems
- GS4.** writing notes, reports, observations, emails
- GS5.** using personnel protective devices
- GS6.** maintaining clean and healthy work environment
- GS7.** using and operating safety devices and equipment
- GS8.** conducting work following workplace security processes and rules
- GS9.** responding to emergency situations pertaining to workplace
- GS10.** understanding people and collaborating to create a healthy workplace

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adhere to standard safety procedures of the organisation</i>	7	10	-	-
PC1. comply with general safety procedures and those for handling equipment, tools, chemicals, and hazardous material, as prescribed and followed in the organisation	-	-	-	-
PC2. remove finger rings or any other metal objects likely to interfere with the work	-	-	-	-
PC3. ensure that identification badge or any other object worn around the neck or on the clothing does not get caught in any rotating machine, or otherwise interfere with the work	-	-	-	-
PC4. use appropriate safety devices such as goggles, gloves, ear plugs, caps, ESD pins, covers, shoes, helmets etc. recommended for the work being performed	-	-	-	-
PC5. inform, escalate, or raise alarm about any suspicions, unaccounted hazardous material, devices, or other objects found in the premises	-	-	-	-
PC6. inform, escalate, or raise alarm about any breach of safety or security procedure in the organisation	-	-	-	-
PC7. help achieve zero accidents goals at work	-	-	-	-
PC8. avoid damage to sensitive electronic components due to negligence of ESD procedures	-	-	-	-
PC9. participate regularly in fire drills or other safety related workshops organised by the organisation	-	-	-	-
PC10. follow strictly all access control and perimeter safety procedures in designated factory areas such as robotic work stations, automated production lines, automated material movement and other potentially risky operations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure that other people follow all access control and perimeter safety procedures in designated factory areas and help avoid accidents	-	-	-	-
PC12. use emergency switches or other mechanisms of stopping a machine immediately in case any emergency situation has developed or about to happen	-	-	-	-
PC13. ensure that electrical equipment are properly grounded	-	-	-	-
PC14. follow Cyber Security guidelines and be vigilant at workplace	-	-	-	-
PC15. proceed to designated safe assembly area immediately on hearing fire alarm	-	-	-	-
<i>Follow healthy practices and posture</i>	8	10	-	-
PC16. wash hands and use sanitizers as recommended to prevent spread of diseases	-	-	-	-
PC17. follow common personal hygiene practices	-	-	-	-
PC18. maintain appropriate posture, especially in long hours of sitting or standing position and in handling heavy materials	-	-	-	-
PC19. participate in company organised health sessions such as exercises, games, yoga, physiotherapy, and other activities	-	-	-	-
PC20. handle heavy and hazardous materials with care, while maintaining appropriate posture, using suitable tools, and handling equipment such as trolleys, jacks, and ladders	-	-	-	-
PC21. learn and apply first aid devices available in the workplace	-	-	-	-
PC22. learn and apply safety and handling procedures for electrical shock and electrocution	-	-	-	-
PC23. learn and apply emergency medical help services	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. follow workplace decorum and avoid emotional outbursts or inappropriate language	-	-	-	-
PC25. prevent any harassment at workplace	-	-	-	-
<i>Practice waste management and recycling</i>	-	-	-	-
PC26. identify recyclable, non-recyclable, and hazardous waste generated in the workplace and comply with their disposal procedures	-	-	-	-
PC27. dispose non-recyclable waste and hazardous waste following recommended processes	-	-	-	-
PC28. deposit recyclable and reusable material at identified locations	-	-	-	-
PC29. support education and compliance of waste management processes	-	-	-	-
<i>Conserve material and resources</i>	-	-	-	-
PC30. identify ways to optimize usage of material and resources such as water, electricity, energy in various tasks, activities, and processes	-	-	-	-
PC31. check for spills and leakages of material in various tasks, activities, and processes and plug them	-	-	-	-
PC32. escalate the leakage issue to appropriate authority if needed	-	-	-	-
PC33. carry out routine cleaning of tools, machines, and equipment and maintain them in good working condition to optimize efficiency and wastage	-	-	-	-
PC34. check if the equipment is functioning normally before commencing work and rectify or report any malfunctioning to the responsible agency	-	-	-	-
PC35. check for any odour, sparks, fumes, emission, unusual vibration, noise, or any other objectionable presence in the environment and take immediate corrective action followed by report to responsible agency	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC36. ensure electrical equipment are properly connected for use and are switched off when not in use	-	-	-	-
PC37. support education and compliance of resource conservation processes	-	-	-	-
NOS Total	15	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0505
NOS Name	Follow health, safety and environment guidelines at workplace
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Defence Equipment, Fire-Fighting & Safety Equipment, Homeland Security
Occupation	Service
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	01/10/2025
Next Review Date	01/10/2030
NSQF Clearance Date	01/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for the Qualification Pack will be created by CGSC.
2. Performance Criteria (PC) have been assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
3. The assessment for the theory part will/may be based on knowledge bank of questions approved CGSC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. Assessment Agencies will create Assessor Guides comprising of Theory and Practical Assessment Set and Guidelines for each examination/training centre (as per assessment criteria below). The same will be approved by CGSC for adequacy.
6. To successfully attain Certification on the Qualification Pack, the trainee must score a minimum of 70% in each Core NOS and minimum of 70% in all non-core NOS. In addition, a candidate needs to attain a minimum overall pass percentage of 70% for certification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CSC/N0221.Oversee the Setup of Automated Metal Cutting Plasma.	40	39	0	21	100	20
CSC/N0222.Select and implement automated CNC plasma cutting techniques for metal fabrication.	40	40	0	20	100	20
CSC/N0223.Post-Cutting Quality Check and Finishing	40	40	0	20	100	20
CSC/N1339.Collaboratively coordinate with the team	30	70	-	-	100	15
CSC/N0505.Follow health, safety and environment guidelines at workplace	15	20	-	-	35	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	185	239	-	61	485	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.